

PROJECT PROGRESS REPORT

Stock Market Prediction using LSTM and RL

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Chapter 1

Project Overview

1.1 Project Title

Stock Market Prediction using LSTM and RL.

1.2 Objective

- To develop a stock market prediction system using LSTM and RL.
- To analyze historical stock market data and identify patterns.
- To predict future stock price movements with improved accuracy.
- To implement an intelligent trading strategy using Reinforcement Learning.
- To evaluate the performance of the predictive and trading models.

1.3 Tools and Technologies

- Python
- NumPy
- Pandas
- Matplotlib
- yfinance
- seaborn

Chapter 2

Weekly Progress Log

Week 1

Tasks Completed

- Developed a working knowledge of core Python syntax
- Covered the fundamentals of multi-dimensional data structures with NumPy.
- Familiarized with data pre-processing and dataframe management using Pandas.
- Gained exposure to data visualization and basic plotting using Matplotlib.

Challenges Faced

- Tackling with multi-dimensional arrays due to its non-intuitive nature.
- Navigating through numerous functions available in pandas that is used for data-processing.
- Understanding the Kwargs and working with multi-chart layouts.

Learning Outcomes

Developed a strong foundational understanding of the core Python libraries widely used in data science and machine learning workflows. Gained proficiency in performing numerical computations and manipulating multi-dimensional arrays using NumPy, cleaning, transforming, and organizing structured datasets with Pandas, and creating meaningful visualizations through Matplotlib. These skills provided the necessary groundwork for efficient data analysis, preprocessing, and interpretation, which are essential components of the stock market prediction project.

Plan for Next Week

- Diving onto the financial aspects and basic terminologies needed to understand stock market.

- Learning about GitHub for future use-cases and projects.

Week 2

Tasks Completed

- I learned to decode market psychology using single and multiple candlestick patterns to identify trends, indecision, and potential price reversals.
- I grasped how to calculate and apply moving averages (SMA/EMA) to identify trend direction and use the MACD and RSI to spot shifting momentum.
- Practiced libraries like yfinance and seaborn for extracting real-world data and plotting of data to further deepen the understanding about the market

Challenges Faced

- Decoding market sentiment across uptrends, downtrends, and sideways phases required intense focus and intricate analysis.
- Completely detaching from a company's fundamental factors to look strictly at technical signals was a significant challenge.

Learning Outcomes

- While I learned that markets behave quite chaotically, adopting a set of fixed principles—with room for flexibility—is crucial for a sustainable, long-term trading career.
- Using yfinance to programmatically apply technical indicators like SMA, EMA, and MACD helped me confidently bridge this gap and execute data-driven strategy validation.

Plan for Next Week

- This practical foundation prepares me perfectly for the coming week, where I will transition into the core fundamentals of Machine Learning to explore data-driven modeling.

Week 3

Tasks Completed

- Studied the fundamental concepts of supervised learning, including regression, classification, and Support Vector Machines (SVMs).
- Explored unsupervised learning techniques with a focus on K-Means clustering for pattern discovery and data segmentation.
- Learned the architecture and working principles of neural networks, including the roles of neurons, activation functions, and layered model structures.

Challenges Faced

- Understanding the mathematical foundations underlying machine learning algorithms, including optimization techniques, loss functions, and gradient-based learning, was particularly challenging despite the intuitive nature of their outcomes.
- Interpreting the relationship between model parameters, training processes, and predictive performance required considerable effort and repeated practice.
- Complex machine learning architectures are often built by combining simpler models and concepts. Understanding how these components interact and contribute to overall model performance proved difficult.
- Selecting appropriate hyperparameters and understanding their impact on model behavior highlighted the importance of careful model tuning.
- Bridging the gap between theoretical concepts and practical implementation required extensive experimentation and debugging to gain a deeper understanding of the learning process.

Learning Outcomes

- Developed a comprehensive understanding of fundamental machine learning paradigms, including supervised and unsupervised learning techniques, and their practical applications in solving real-world problems.
- Strengthened my understanding of machine learning model architectures and the principles governing their training, evaluation, and optimization.
- Gained deeper insight into the mathematical concepts underlying machine learning algorithms, including loss functions, gradient descent, and model performance metrics.

- Learned to leverage data visualization techniques to better interpret model behavior, decision boundaries, clustering patterns, and learning processes.
- Established a strong foundation in neural network concepts, preparing for the study of advanced deep learning architectures such as CNNs, RNNs, LSTMs, and Reinforcement Learning models.

Plan for Next Week

- Moving forward, I will advance to more complex architectures, focusing on Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Reinforcement Learning (RL).

Chapter 3

Final Summary

3.1 Overall Progress

The project is nearly halfway complete, and I have developed a strong understanding of both the theoretical concepts and practical implementation required for building the final model. I plan to leverage these skills to improve and optimize the performance of the algorithm that I will further build.